Ogbonna Arinze

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SUMMARY

Senior Frontend Engineer with 8 years of experience architecting and delivering high-quality, user-centric web applications across healthcare, fintech, and enterprise B2B environments. Deep expertise in **React**, **TypeScript**, and modern **JavaScript**, with a track record of building reusable component libraries and design systems that establish company-wide UI standards. Skilled in frontend performance optimization, automated testing (**Jest**, **Playwright**, **Cypress**), **CI/CD pipelines**, and **web security** (JWT, OAuth2, RBAC), with hands-on experience in real-time data streaming, reactive state management, and AI-assisted development workflows.

CORE COMPETENCIES

- **Frontend Architecture:** React, TypeScript, Next.js, Vue.js, reusable design systems (Radix UI), scalable component libraries, SSR/SSG, and monorepos (Nx, Turborepo).
- **State Management:** Redux, Redux Toolkit, React Query, Zustand, RxJS (reactive data streaming), and Axios.
- **Build Tools & Performance:** Webpack, Rollup, Vite, code splitting, lazy loading, bundle analysis (webpack-bundle-analyzer, rollup-plugin-visualizer), Lighthouse, and Core Web Vitals.
- **Testing & Quality:** Jest, Enzyme, Playwright, Cypress, Allure Reports, Storybook, ESLint, Prettier, Husky, Lint-staged, and CI-driven QA.
- **Web Security:** JWT-based authentication, OAuth2, role-based access control (RBAC), HTTP-only cookies, secure token refresh, and permission-based UI rendering.
- **CI/CD & Infrastructure:** GitHub Actions, Docker, AWS (CloudFront, S3), Vercel, automated deployment pipelines, and Nx monorepo orchestration.
- **Real-Time Systems:** Server-Sent Events (SSE), EventSource API, WebSocket patterns, and reactive data streaming with RxJS.
- **AI-Assisted Development:** OpenAI API, LangChain, RAG systems, Vercel AI SDK, vector search (Chroma, pgvector), and agentic AI workflows.
- **Leadership & Process:** Kanban, Agile/Scrum, technical architecture reviews, mentorship, code reviews, TDD/BDD, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, establishing company-wide UI standards that improved consistency, **accessibility (WCAG)**, and developer velocity across multiple product teams.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing user adoption.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering across protected routes.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools**.
- Introduced **Redux** for global state management and **React Query** for server-state caching, refactoring legacy class components to functional components and streamlining reactive data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, demonstrating hands-on AI-assisted development capabilities.
- Mentored a team of two junior developers (interns), conducting code reviews and fostering a culture of technical excellence and high engineering standards.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.

- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improved code reliability.
- Utilized **Storybook** for documenting **React** components, streamlining change management and ensuring clear communication of UI component APIs across the team.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating **CI/CD** processes across the development lifecycle.
- Mentored junior engineers on frontend best practices and conducted peer code reviews, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%** through optimized transaction processing interfaces.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Adopted the native browser **EventSource API** for **real-time data streaming**, replacing conventional socket.io methods to achieve faster data updates and a more responsive interface for live transaction monitoring.
- Built a real-time account reconciliation solution, boosting daily reconciliation throughput by **60%** through innovative transaction processing and data management strategies.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI components.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance, code reviews, and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, and devised comprehensive documentation for each application detailing all functions, capabilities, and features.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: NestJS API backend, **Next.js** citizen-facing web app with **22 interactive chart types** (Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring auto-generated data visualizations from AI responses, dark mode, URL state management, and responsive mobile layouts.
- Built a **multi-agent AI system** using the Mastra framework with 5 specialized agents and a hybrid **RAG pipeline** combining pgvector similarity search with BM25 full-text search, Reciprocal Rank Fusion, and Cohere reranking.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*